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# Analog MEMS Electronic Stethoscope

Literature Review and Theoretical Analysis

Prathishanth. A 220480P  
Dayananthan.T 220096T  
Ananthakumar.T 220029T

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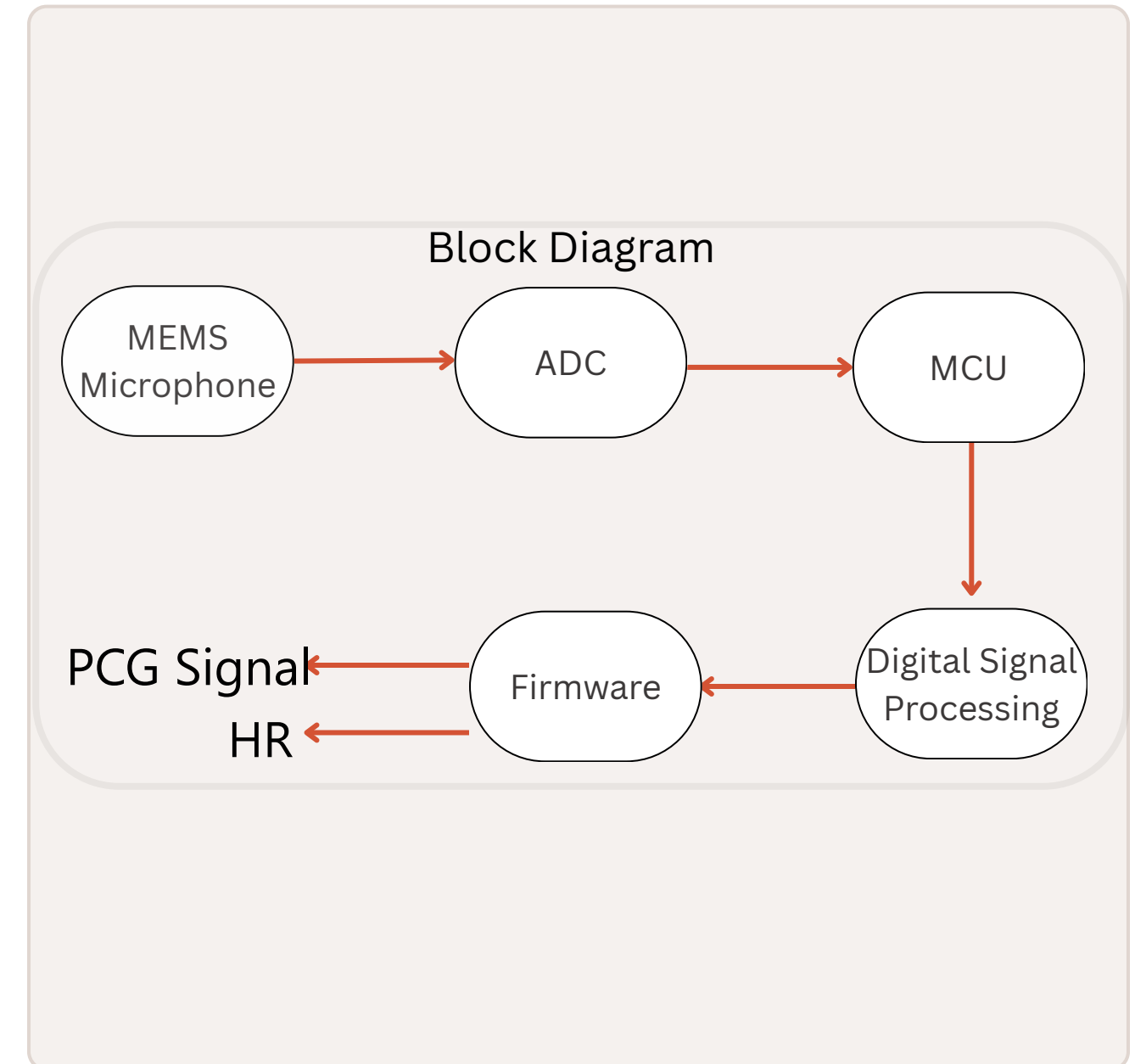
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# Introduction

- Heart sounds are acoustic vibrations generated by valve closure and turbulent blood flow during the cardiac cycle, mainly identified as S1 (“lub”) and S2 (“dub”).
- Conventional stethoscope auscultation is a low-cost primary cardiac assessment method but depends on clinician experience and hearing sensitivity.
- Electronic stethoscopes convert heart sounds into electrical signals, enabling amplification, filtering, recording, and digital processing for more objective cardiac screening.



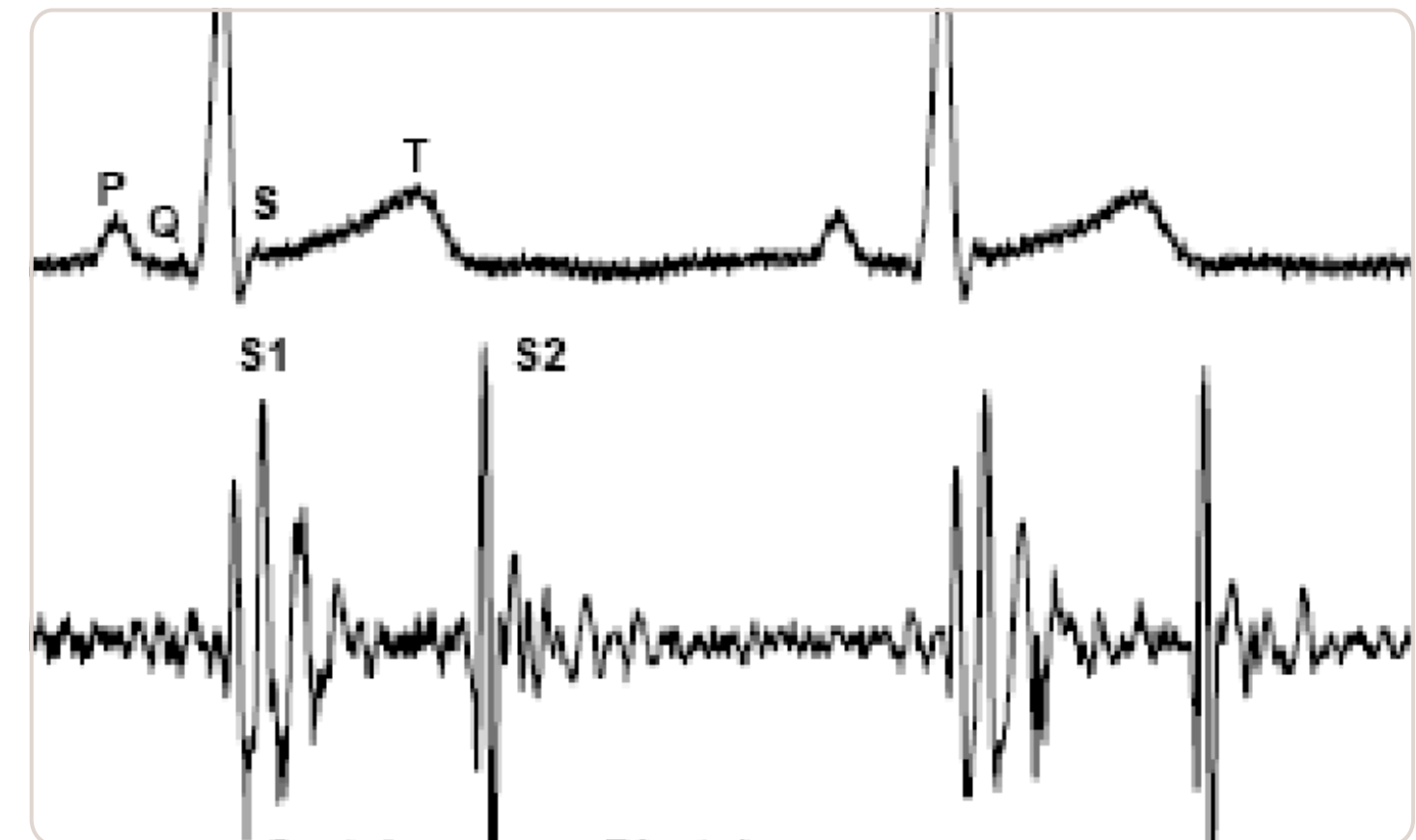
**Transition:** Before studying filtering and processing methods, it is first necessary to understand how heart sounds are generated and how they are represented as a PCG signal.



# Heart-Sound Generation

- **S1 (“lub”)**: produced by closure of the mitral and tricuspid valves at the onset of systole; marks the start of ventricular contraction.
- **S2 (“dub”)**: produced by closure of the aortic (A2) and pulmonary (P2) valves at the end of systole; A2 normally precedes P2, giving a physiological “split” typically < 30–40 ms.
- **S3 / S4**: lower-amplitude sounds linked to rapid ventricular filling (S3) or atrial contraction against a stiff ventricle (S4); often physiological in youth but can indicate pathology in adults.
- **Timing**: S1 aligns with the QRS complex of the ECG; S2 follows the systolic pause — confirming the well-known ECG–PCG synchrony.
- **Frequency content**: S1  $\approx$  10–200 Hz; S2  $\approx$  50–300 Hz

Cardiac cycle with ECG and heart sounds aligned

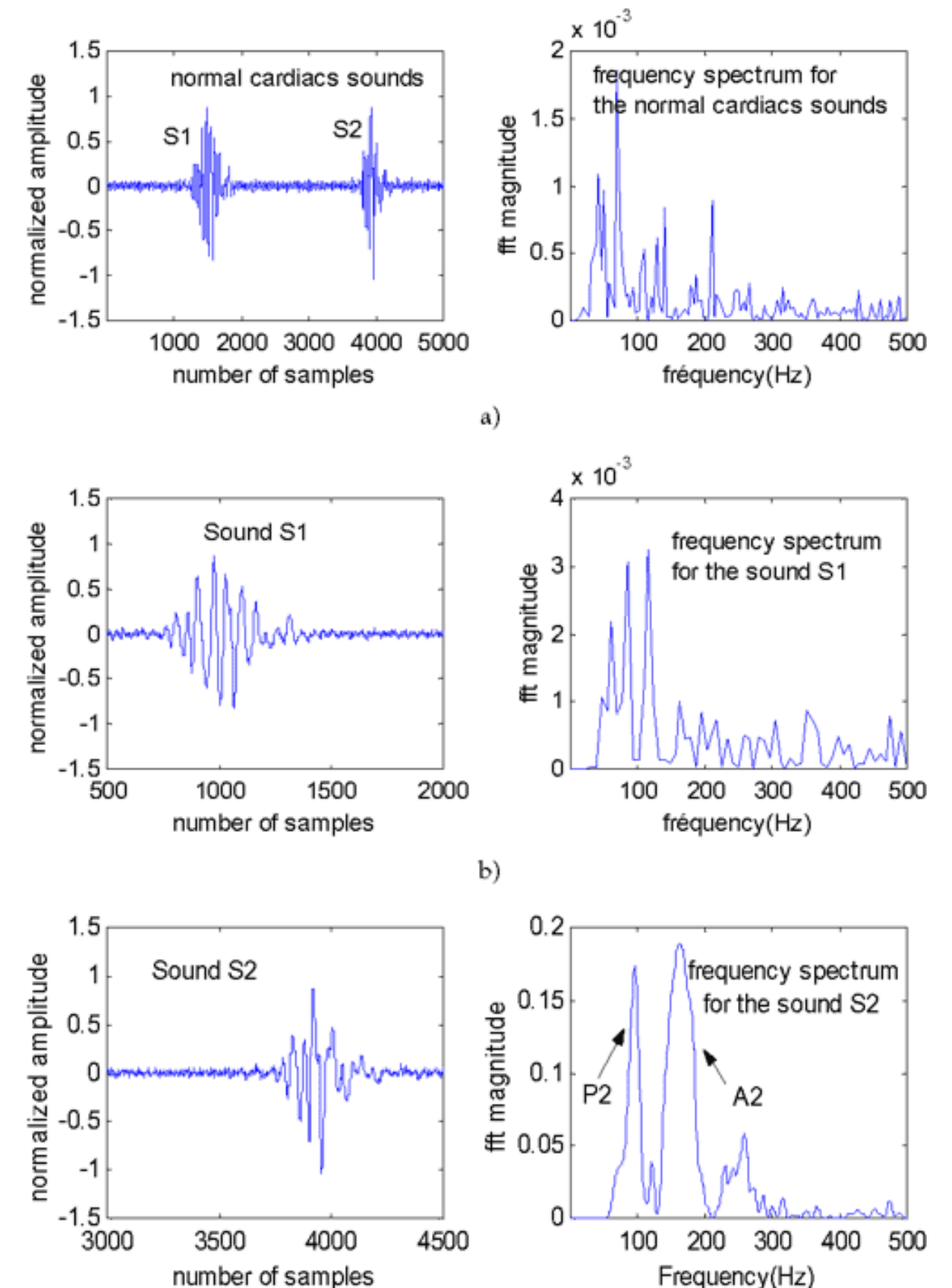




# Phonocardiogram (PCG) Fundamentals

- **Definition:** a PCG is the electronically recorded, time-domain waveform of heart sounds acquired via a microphone or acoustic sensor placed on the chest.
- **Relation to auscultation:** PCG objectifies what a clinician hears — it converts a subjective, transient auditory event into a permanent, quantifiable, reviewable signal.
- **Waveform characteristics:** each cardiac cycle appears as two short, high-energy bursts (S1, S2) separated by relatively quiet systolic and diastolic intervals.
- **Cardiac-cycle labelling:** each cycle is segmented and labelled S1 – systole – S2 – diastole, forming the basis for beat-by-beat feature extraction.
- **Clinical importance:** PCG enables objective, storable, and computer-analysable cardiac screening, supporting early detection of pre-existing heart disease in low-resource settings.

$$HR = 60 / (TS1(n+1) - TS1(n))$$



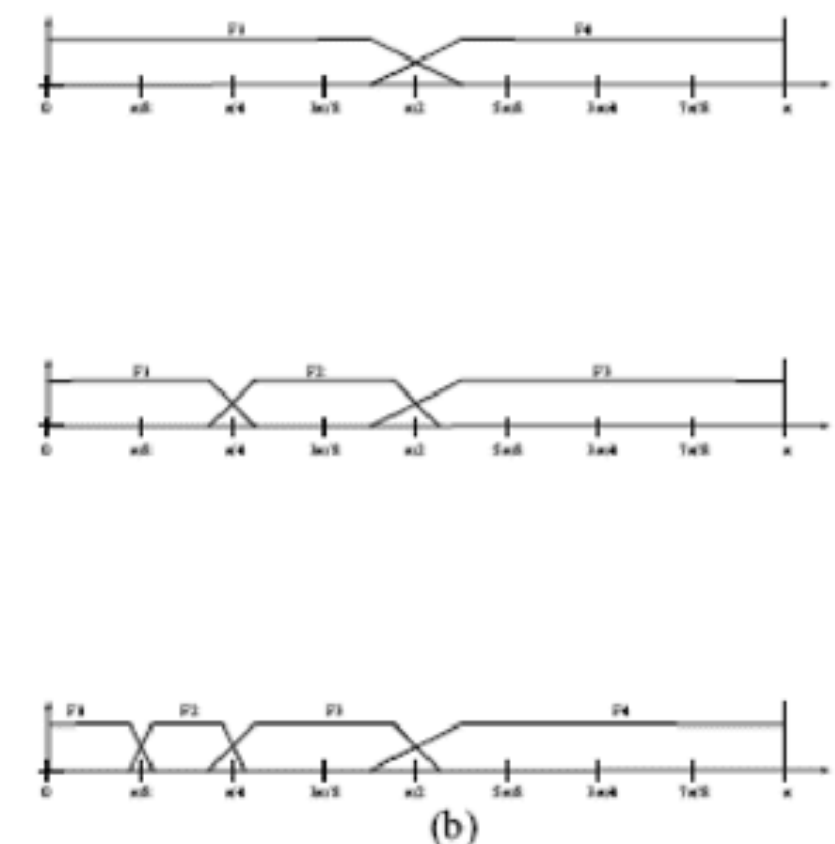
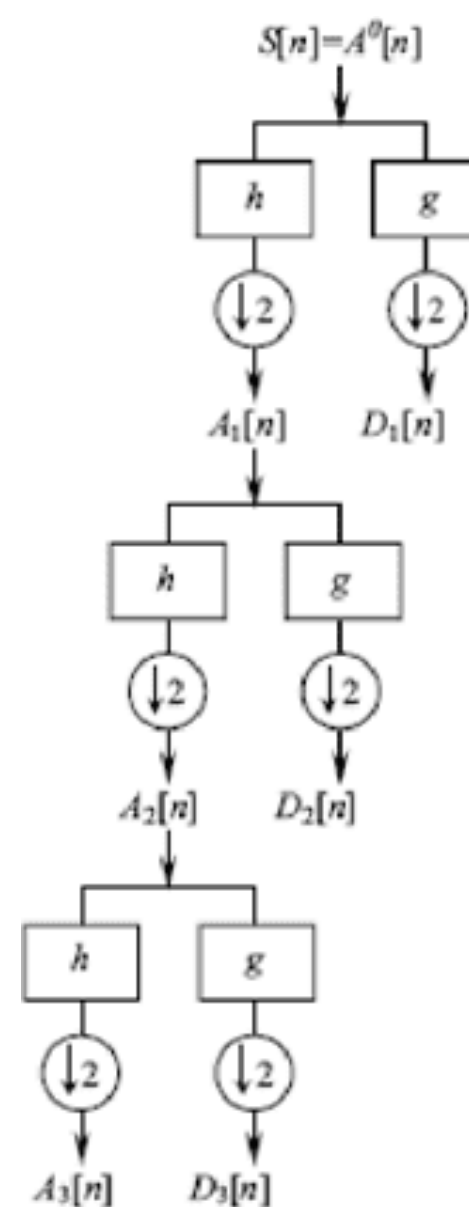


# Signal Characteristics Relevant to Digital Processing

- **Frequency spectrum:** heart sounds are predominantly low-frequency, roughly 20–200/300 Hz (S1: 10–200 Hz, S2: 50–300 Hz), well within the audio band but far below typical ambient/electronic noise bandwidths.
- **Key noise sources:**
  - Lung/breath sounds
  - Environmental/ambient noise.
  - Power-line interference
- **Why filtering is necessary:** raw acquired signals are dominated by out-of-band and non-stationary noise; band-pass and adaptive filtering isolate the diagnostically relevant heart-sound band and improve SNR before any feature extraction.
- **Sampling frequency:** per the Nyquist criterion,  $f_s \geq 2 \times f_{\max}$ ; PCG is commonly sampled at 2–8 kHz (e.g., 8 kHz acquisition, downsampled to 2 kHz) — comfortably above the heart-sound bandwidth.
- **Preparation for feature extraction:** a clean, correctly-sampled digital PCG is the prerequisite for downstream tasks such as S1/S2 detection, heart-rate estimation, and normal/pathological classification.

**Sampling theorem:**  $f_s \geq 2 \times f_{\max}$

With heart-sound content below ~300 Hz, sampling at 2–8 kHz provides ample margin above Nyquist for anti-alias-safe acquisition.



# What is an Electronic Stethoscope?

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- An electronic stethoscope converts acoustic heart and lung sounds into electrical signals, enabling amplification, filtering, recording, and digital analysis for improved diagnosis.
- The output is a phonocardiogram(PCG)

## Limitations of Acoustic Stethoscopes

- Fixed mechanical gain
- No reproducibility
- Total dependence on ambient quietness

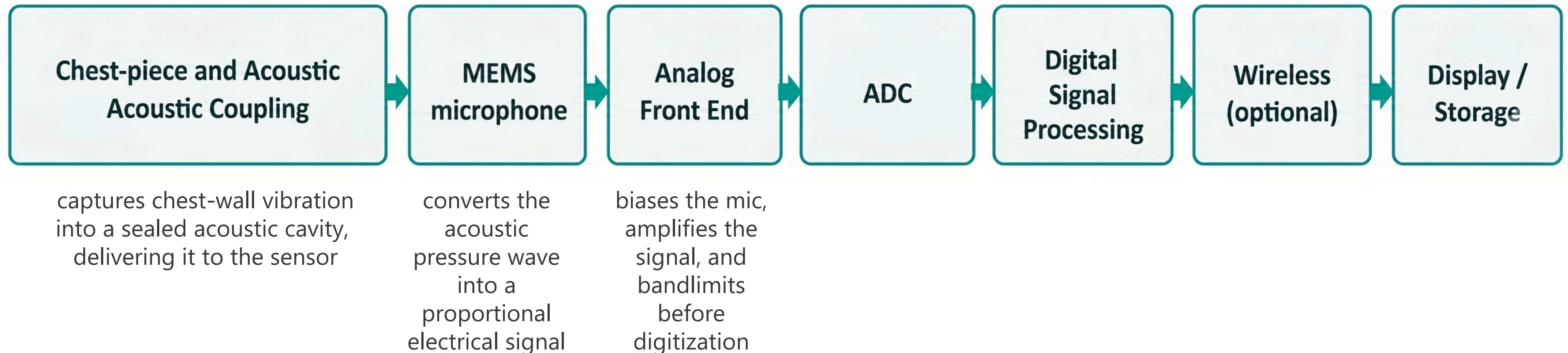


# Operating Principle of Electronic stethoscope

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- Body sounds → Transducer → Amplifier → Filter → ADC → Digital Signal Processing → Output
- Transducer types:
  - Micro-Electro-Mechanical Systems(MEMS) microphones
  - Electret Condenser Microphone(ECM)
  - Piezoelectric sensor

## Complete System Block Diagram



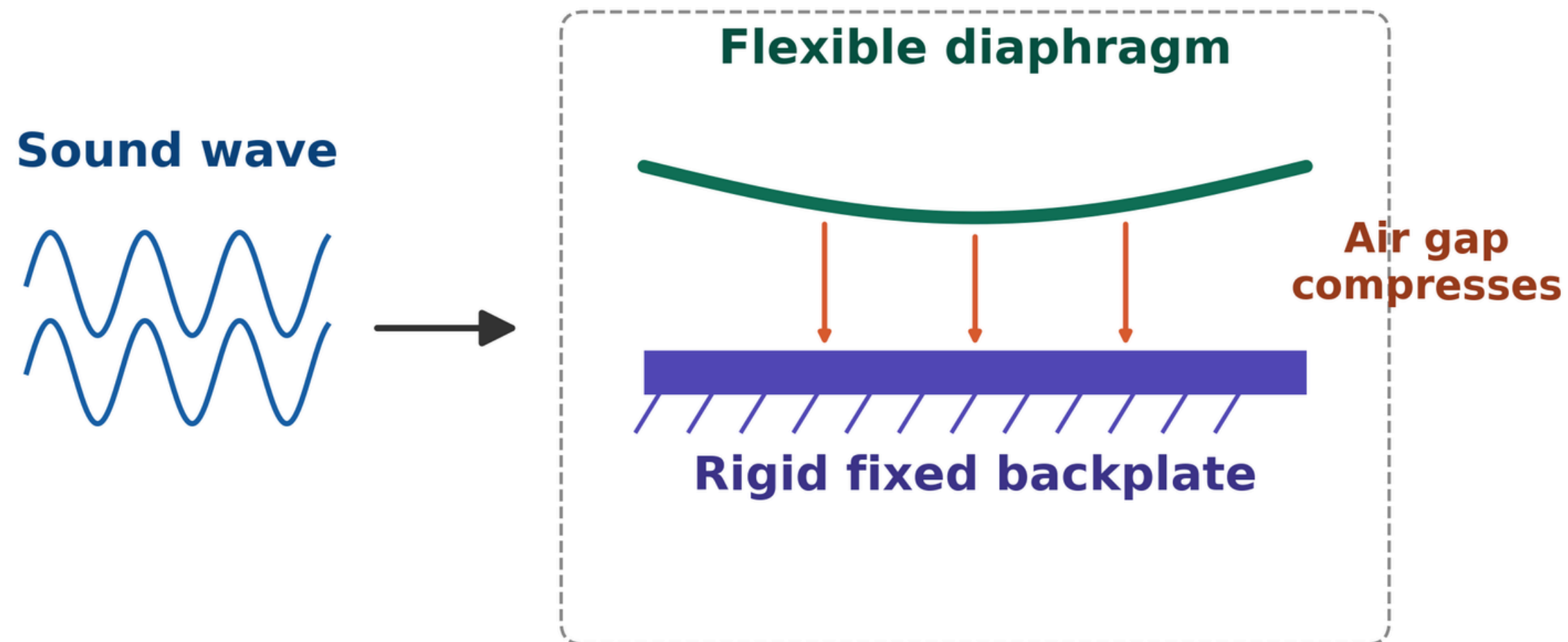
# Literature Review(Hardware)

| Paper(Year)                                   | Microphone Model          | SNR sensitivity      | Chest piece Design                         | MCU               | Onboard Processing Capability    |
|-----------------------------------------------|---------------------------|----------------------|--------------------------------------------|-------------------|----------------------------------|
| Anumukonda et al. (2015)                      | ADMP441<br>Digital MEMS   | 61 dBA<br>26 dBFS    | Custom / Simple setup                      | ARM Cortex-M3     | Limited                          |
| Lee et al. (2021)<br>Finger-ring hybrid       | ICS-43434<br>Digital MEMS | 64–65 dBA<br>26 dBFS | Fully Custom                               | EFR32MG + BLE SoC | Strong Digital filter            |
| Goh et al. (2023)<br>3D-printed low-cost      | SEN0487<br>Analog MEMS    | 59 dBA<br>42 dB      | 3D-printed ABS chest-piece + PET diaphragm | Not detailed      | Minimal                          |
| Park et al. (2022)<br>Mouse-shaped multimodal | ICS-43434<br>Digital MEMS | 64–65 dBA<br>26 dBFS | Custom + Commercial diaphragm              | EFR32BG + BLE SoC | real-time pre-processing on MCU. |

# MEMS Microphone: Structure and Working Principle

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- **Micro-Electro-Mechanical System (MEMS) mic** : acoustic transducer fabricated on silicon using semiconductor processes.
- **Structure**: Flexible silicon diaphragm suspended over a fixed backplate → **forms a parallel-plate capacitor**



# Key Specifications for MEMS Microphone Selection

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| Specification       | Preference          |
|---------------------|---------------------|
| Sensitivity (mV/Pa) | Higher              |
| SNR                 | As high as possible |
| Noise floor         | As low as possible  |
| Frequency Response  | Flat                |

# Need for Amplification and Audio Amplification

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- MEMS mic analog output  $\approx$  few mV
- Typical ADC requires input in the 1–3.3 V range for good resolution.
- Raw signal too weak and noisy to digitize directly → **amplification stage mandatory**
- **Low-noise preamplifier** : Instrumentation amplifier amp boosts signal with minimal added noise

# References

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- [1] A. Herath et al., "A Simultaneous ECG-PCG Acquisition System with Real-Time Burst-Adaptive Noise Cancellation," arXiv preprint, arXiv:2510.23819v2, Feb. 2026.
- [2] S. M. A. Debbal and F. Bereksi-Reguig, "Features for Heartbeat Sound Signal Normal and Pathological," Recent Patents on Computer Science, vol. 1, pp. 1–8, 2008.

**THANK YOU**